

Lab Sheet 3

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice SQL Functions.

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: SQL Functions

DATABASE

```
CREATE DATABASE IF NOT EXISTS LabSheet3;
```

```
USE LabSheet3;
```

```
DROP TABLE IF EXISTS Employees;
```

```
CREATE TABLE Employees (  
    Employee_ID INT PRIMARY KEY,  
    First_Name VARCHAR(50),  
    Last_Name VARCHAR(50),  
    Salary DECIMAL(10,2)
```

```
);
```

```
INSERT INTO Employees  
(Employee_ID, First_Name, Last_Name, Salary)  
VALUES
```

```
(1, 'Divya', 'Verma', 45678.45),  
(2, 'Aarav', 'Sharma', 52345.67),  
(3, 'Priya', 'Singh', 38999.32),  
(4, 'Rahul', 'Kumar', 61250.75),  
(5, 'Ananya', 'Gupta', 47890.55);
```

A. Number Functions

1. Demonstrate ABS() with suitable examples.

```
SELECT ABS(-25) AS Absolute_Value;
```

Result Grid				Filter Rows:	
	Absolute_Value				
	25				

2. Demonstrate ROUND() with suitable examples.

```
SELECT ROUND(125.678, 2) AS Rounded_Value;
```

3. Demonstrate CEIL() and FLOOR().

```
SELECT  
CEIL(12.3) AS Ceiling_Value,  
FLOOR(12.9) AS Floor_Value;
```

Result Grid	Filter Rows:	Exports	Wra
Remainder			
2			

4. Demonstrate MOD() with suitable examples.

FROM Employees;

SELECT MOD(17, 5) AS Remainder;

5. Calculate the rounded salary of employees.

SELECT

Employee_ID,

First_Name,

Last_Name,

Salary,

ROUND(Salary, 2) AS Rounded_Salary

B. Aggregate Functions

Find the total number of students using COUNT().

FROM Employees;

SELECT COUNT(*) AS Total_Students

Result Grid	Filter Rows:
Total_Students	
1	

6. Find the total salary of all employees using SUM().

7. Find the average salary using AVG().

FROM Student;

SELECT ROUND(AVG(Salary), 2) AS Average_Salary

8. Find the highest salary using MAX().

FROM Employees;

SELECT MAX(Salary) AS Highest_Salary

9. Find the lowest salary using MIN().

FROM Employees;

SELECT MIN(Salary) AS Lowest_Salary

C. Character Functions

10. Demonstrate UPPER() on student names.

FROM Employees;

SELECT

Student_Name,

Upper(Student_Name)AS Upper_Name

11. Demonstrate LOWER() on student names.

FROM Student;

SELECT

Student_Name,

LOWER(Student_Name) AS Lower_Name

12. Demonstrate LENGTH() on student names.

FROM Student;

SELECT

Student_Name,

LENGTH(Student_Name) AS Name_Length

13. Demonstrate CONCAT() using first and last names

FROM Student;

SELECT

First_Name,

Last_Name,

CONCAT(First_Name, ' ', Last_Name) AS Full_Name.

14. Demonstrate SUBSTRING()/SUBSTR() with suitable examples

FROM Employees;

SELECT

Student_Name,

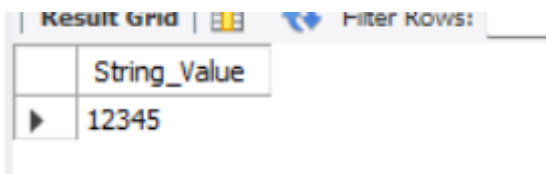
SUBSTRING(Student_Name, 1, 3) AS First_Three_Characters.

D. Conversion Functions

15. Convert a number into a character/string value.

FROM Student;

SELECT CAST(12345 AS CHAR) AS String_Value;

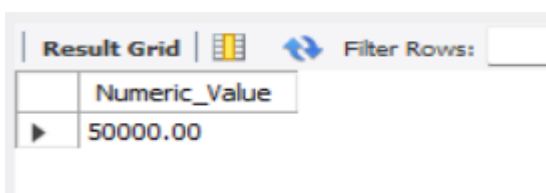


The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid has two columns: 'String_Value' and a value '12345'.

	String_Value
▶	12345

16. Convert a string into a numeric value.

SELECT CAST('50000' AS DECIMAL(10,2)) AS Numeric_Value;



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid has two columns: 'Numeric_Value' and a value '50000.00'.

	Numeric_Value
▶	50000.00

17. Convert a date value into a different format.

SELECT

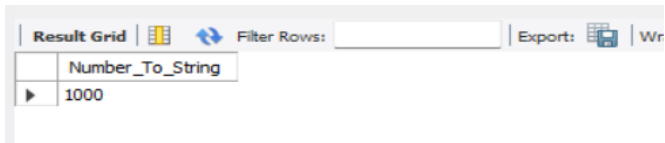
Joining_Date,

DATE_FORMAT(Joining_Date, '%d-%m-%Y') AS Formatted_Date

18. Demonstrate CAST() with suitable examples.

FROM Employees;

SELECT CAST(1000 AS CHAR) AS Number_To_String;



The screenshot shows a 'Result Grid' window with a toolbar containing icons for grid view, refresh, filter, export, and print. Below the toolbar, there is a table with one column named 'Number_To_String' and one row containing the value '1000'.

Number_To_String
1000

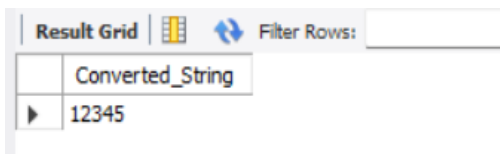
SELECT CAST('2500' AS SIGNED) AS String_To_Number;

SELECT CAST('45678.75' AS DECIMAL(10,2)) AS Decimal_Value;

SELECT CAST('2026-09-23' AS DATETIME) AS Date_Time_Value;

19. Demonstrate CONVERT() where supported by the DBMS.

SELECT CONVERT(12345, CHAR) AS Converted_String;



The screenshot shows a 'Result Grid' window with a toolbar containing icons for grid view, refresh, filter, export, and print. Below the toolbar, there is a table with one column named 'Converted_String' and one row containing the value '12345'.

Converted_String
12345

SELECT CONVERT('5000', UNSIGNED) AS Converted_Number;

SELECT CONVERT('2026-09-23', DATETIME) AS Converted_Date;

E. Date Functions

20. Display the current date using a suitable date function.

SELECT CURDATE() AS Current_Date;

SELECT CURRENT_DATE() AS Current_Date;

SELECT NOW() AS Current_Date_Time;

21. Extract the year, month and day from a date.

SELECT

Joining_Date,

YEAR(Joining_Date) AS Joining_Year,

MONTH(Joining_Date) AS Joining_Month,

DAY(Joining_Date) AS Joining_Day

22. Find the difference between two dates.

SELECT DATEDIFF('2026-09-29', '2026-09-01') AS Date_Difference;

23. Add a specified number of days to a date.

SELECT

Joining_Date,

YEAR(Joining_Date) AS Joining_Year,

MONTH(Joining_Date) AS Joining_Month,

DAY(Joining_Date) AS Joining_Day

FROM Employees;

24. Display employees' joining dates in a required date format.

SELECT

Joining_Date,

```
EXTRACT(YEAR FROM Joining_Date) AS Year,  
EXTRACT(MONTH FROM Joining_Date) AS Month,  
EXTRACT(DAY FROM Joining_Date) AS Day  
FROM Employees;
```

